

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

NET205

Why is it always DNS? Ask us anything

Steve Seymour

WW SA Tech Leader, Networking
AWS

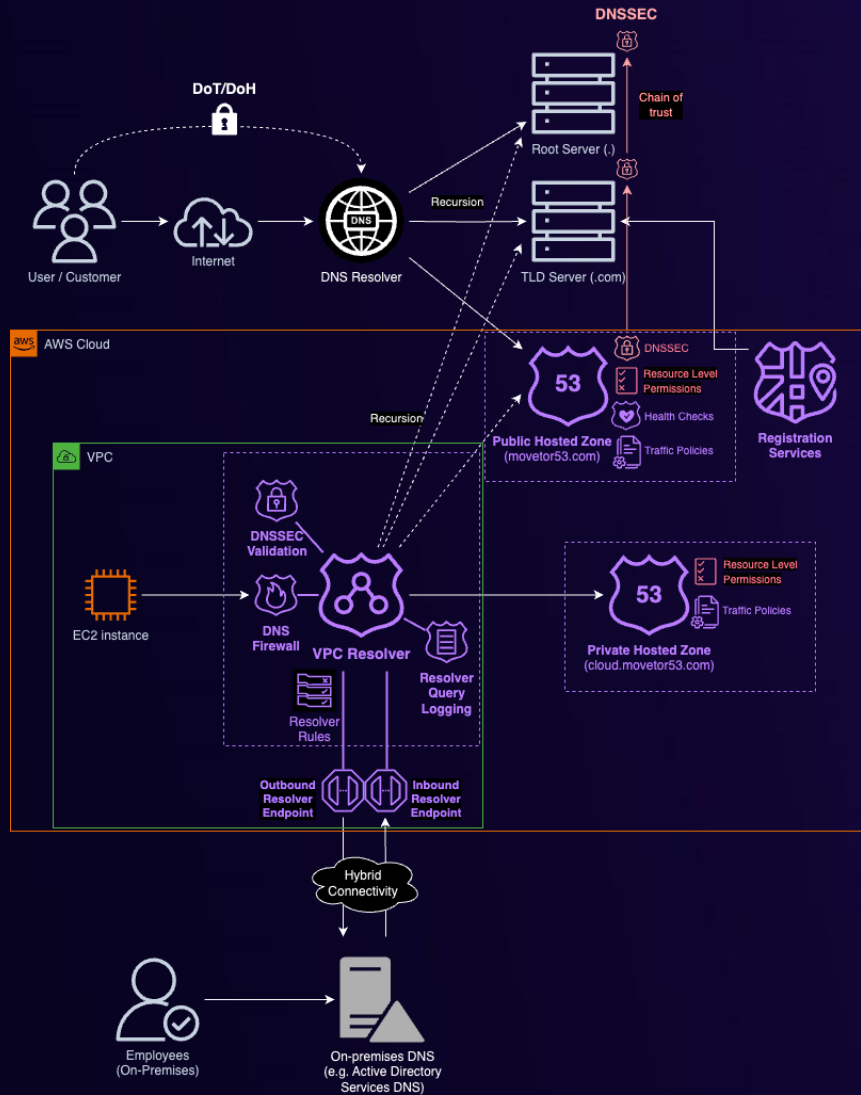
Christian Elsen

Principal Networking SA
AWS



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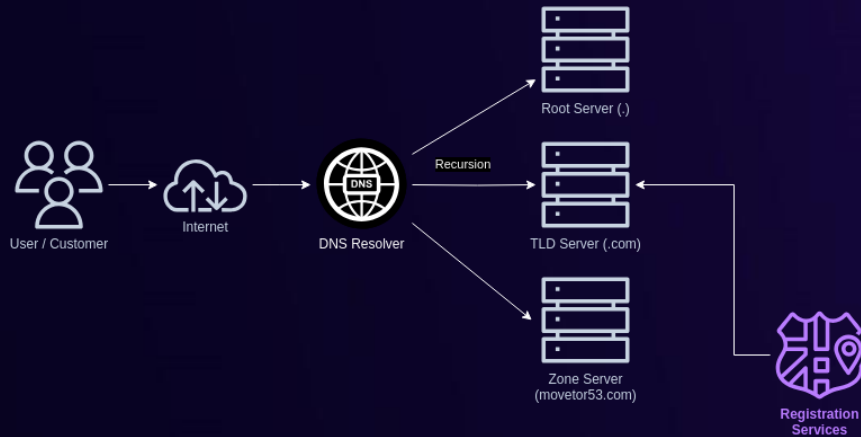
Amazon Route 53: So many use cases



Amazon Route 53 use cases

1. Registration services
2. Public hosted zone
 - Health checks
 - Traffic flow
3. VPC Resolver
4. Private hosted zone
5. Inbound Resolver endpoint
6. Outbound Resolver endpoint
7. Resolver DNS Firewall
8. DNSSEC
9. Query logging

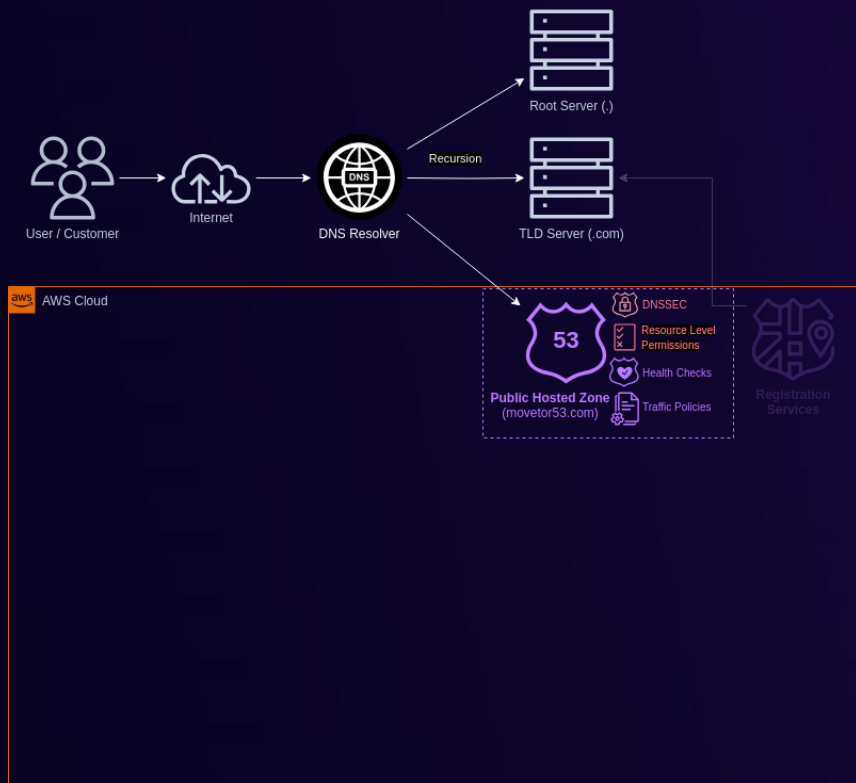
Registration services



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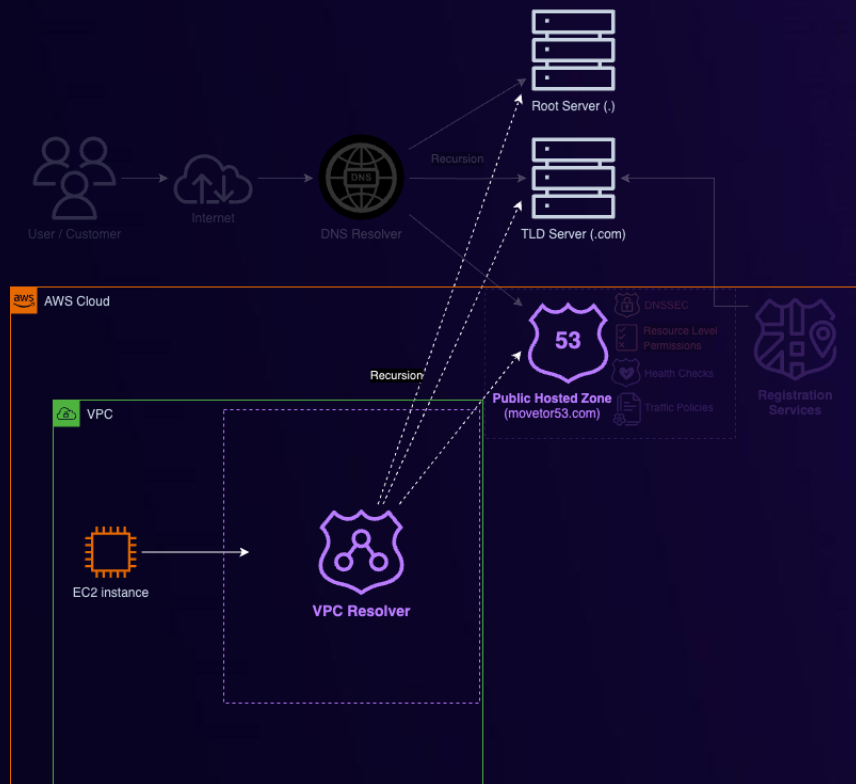
Public hosted zone



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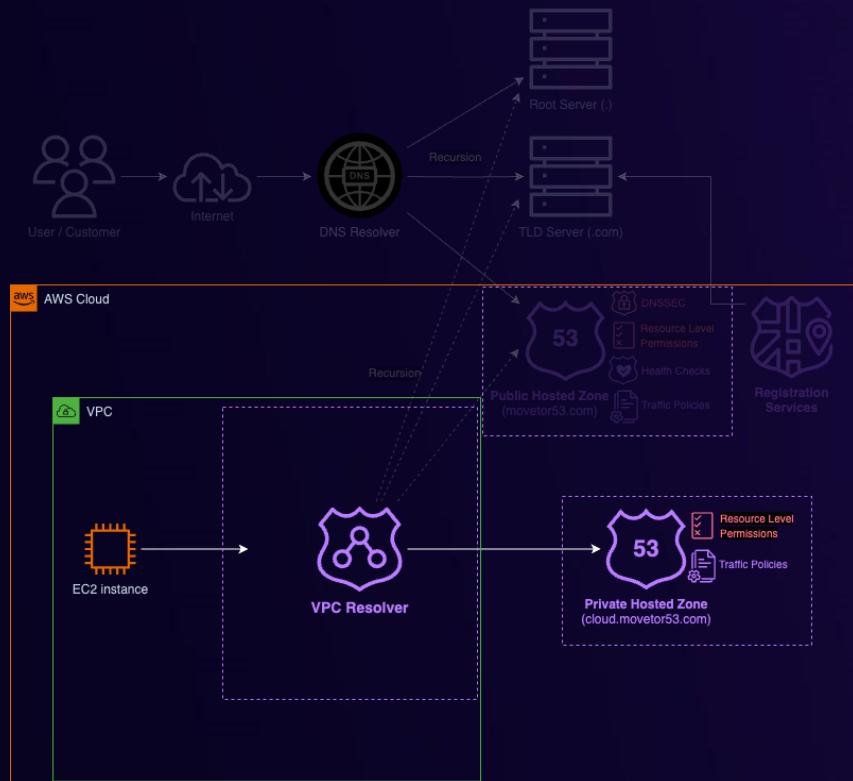
VPC Resolver



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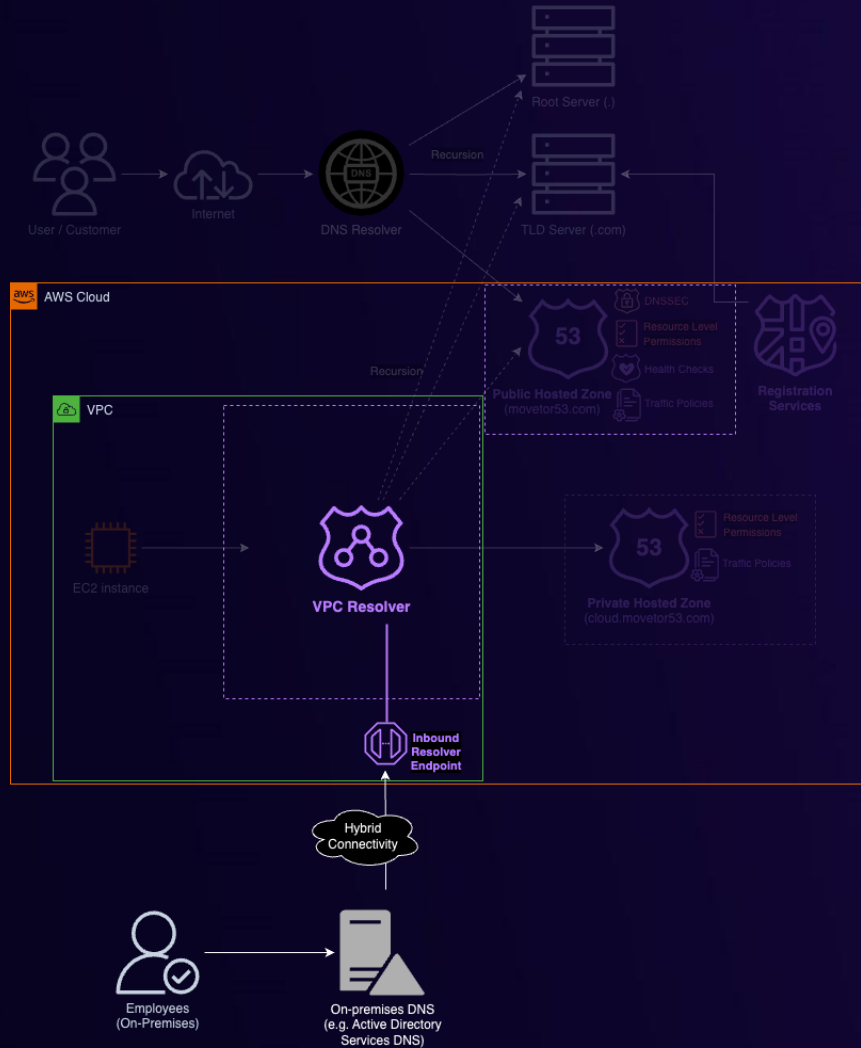
Private hosted zone



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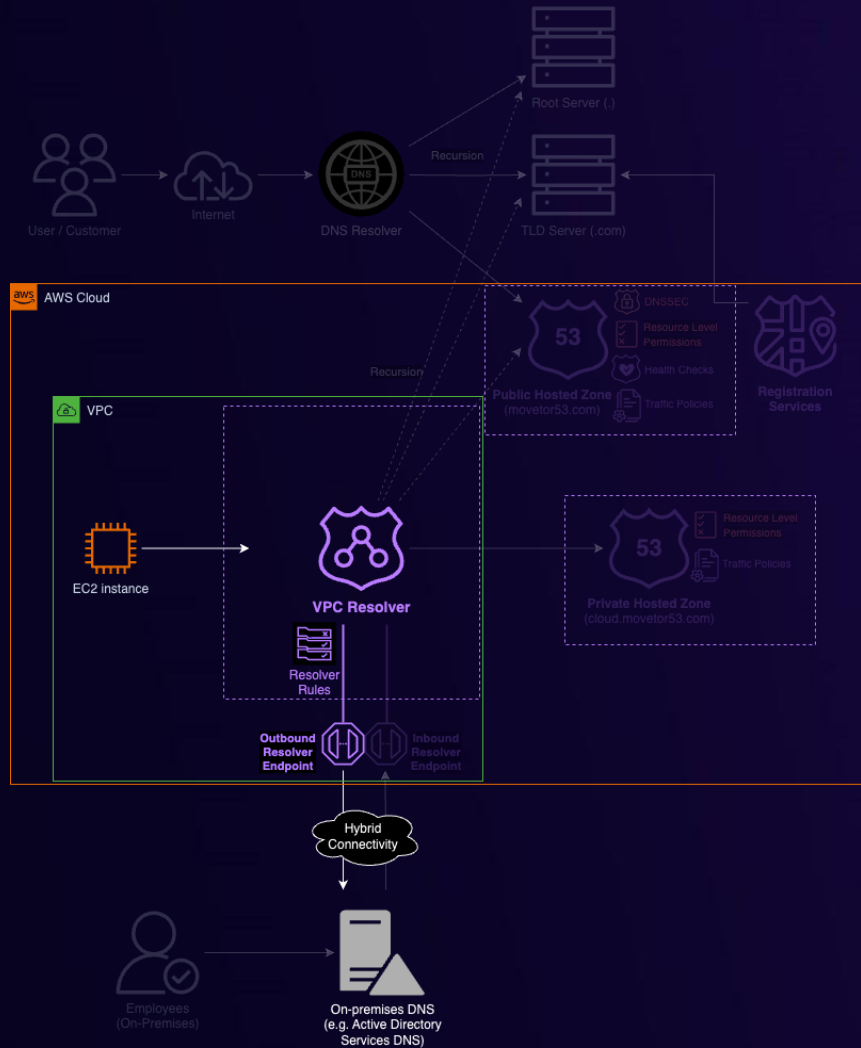
Inbound Resolver endpoint



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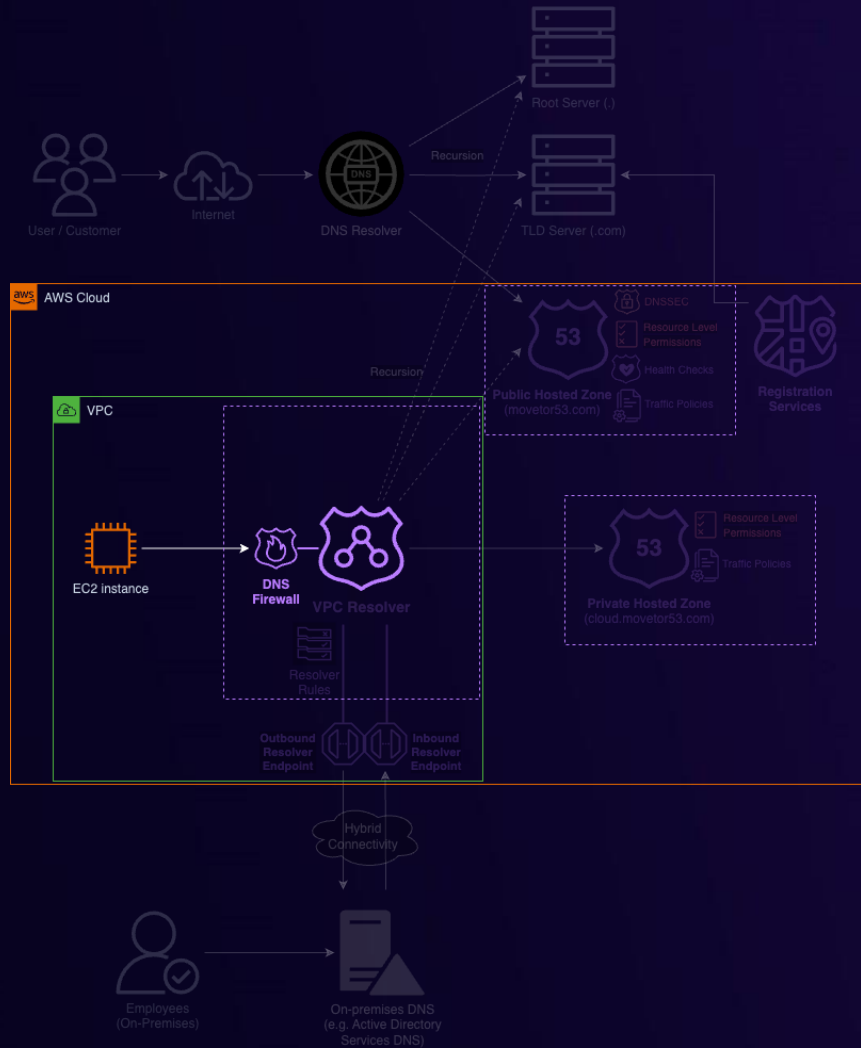
Outbound Resolver endpoint



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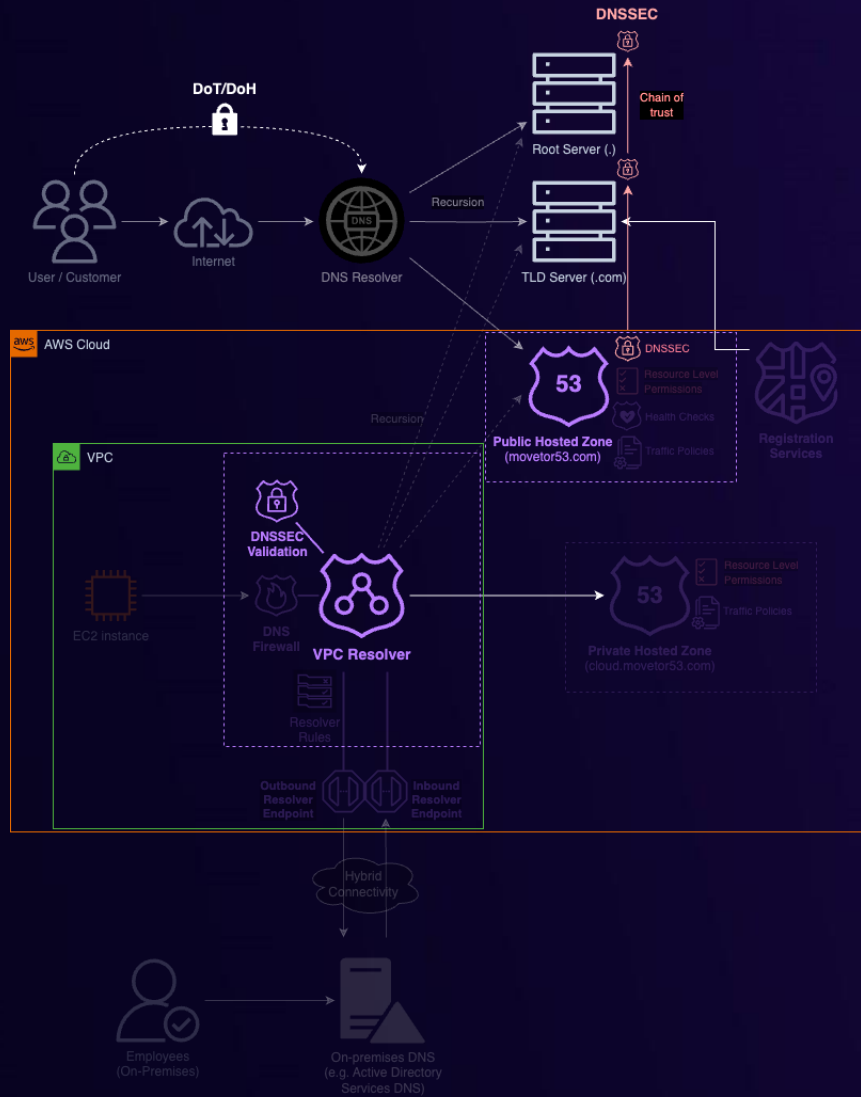
Resolver DNS Firewall



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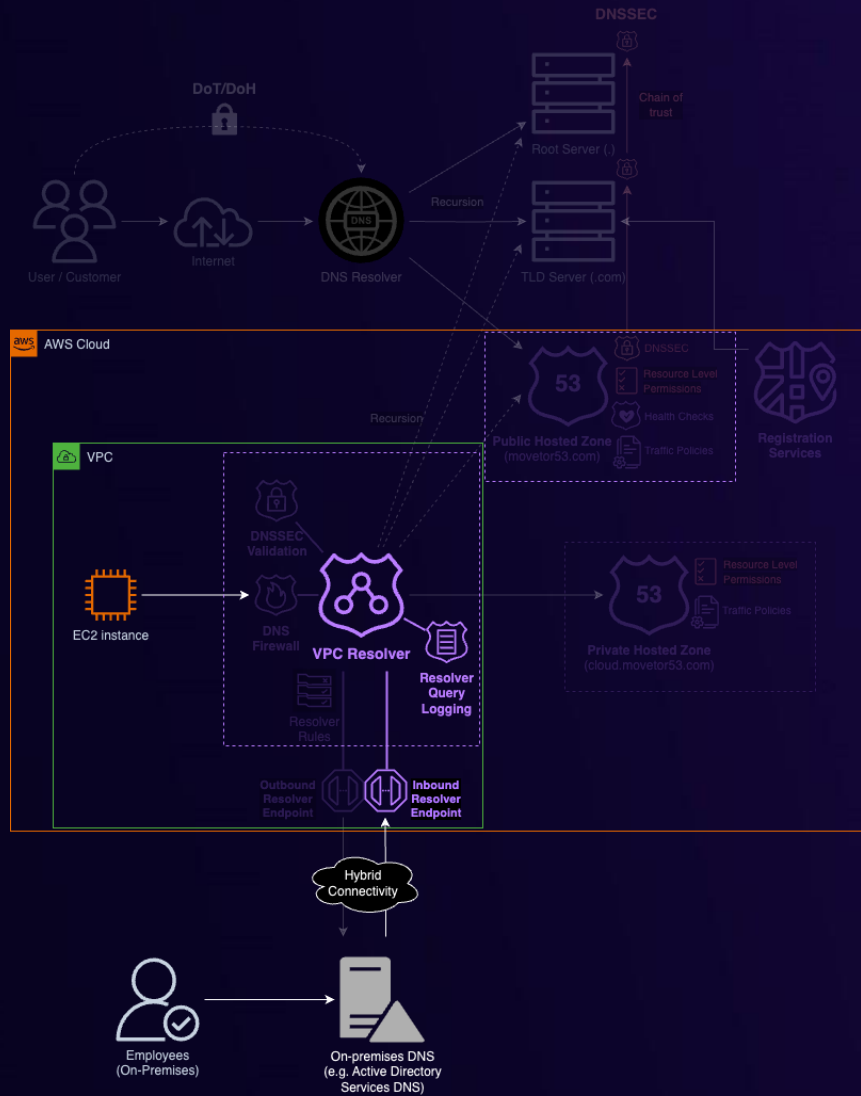
DNSSEC



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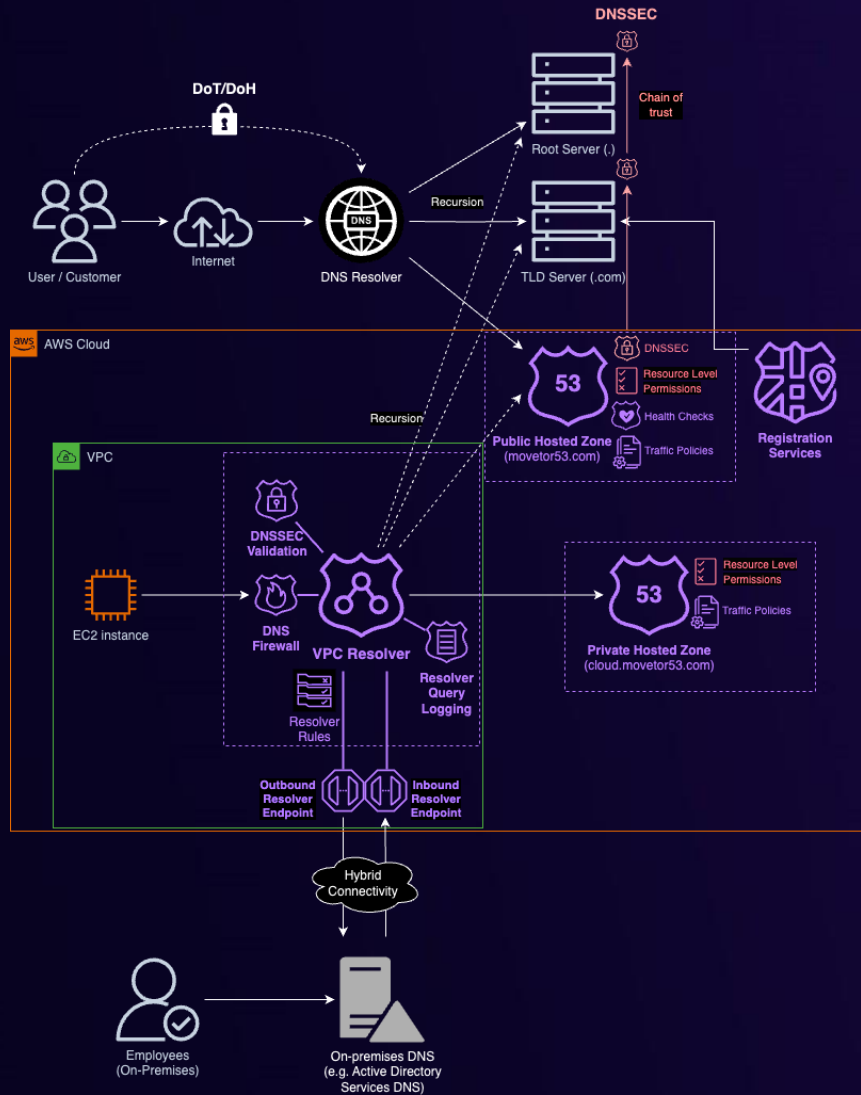
Query logging



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Let's discuss

Use case 1: Migrate a domain registration to AWS or register a new one

Use case 2 : Host a static website using Amazon S3 and Amazon CloudFront

Use case 3: Resolve DNS entries from your EC2 instances in a VPC

Use case 4: Use a domain "cloud.movetor53.com" inside your VPCs

Use case 5: Have on-premises infrastructure resolve private hosted zone entries

Use case 6: Enable Amazon EC2 to look up ADS zone "ads.movetor53.com"

Use case 7: Prevent data exfiltration

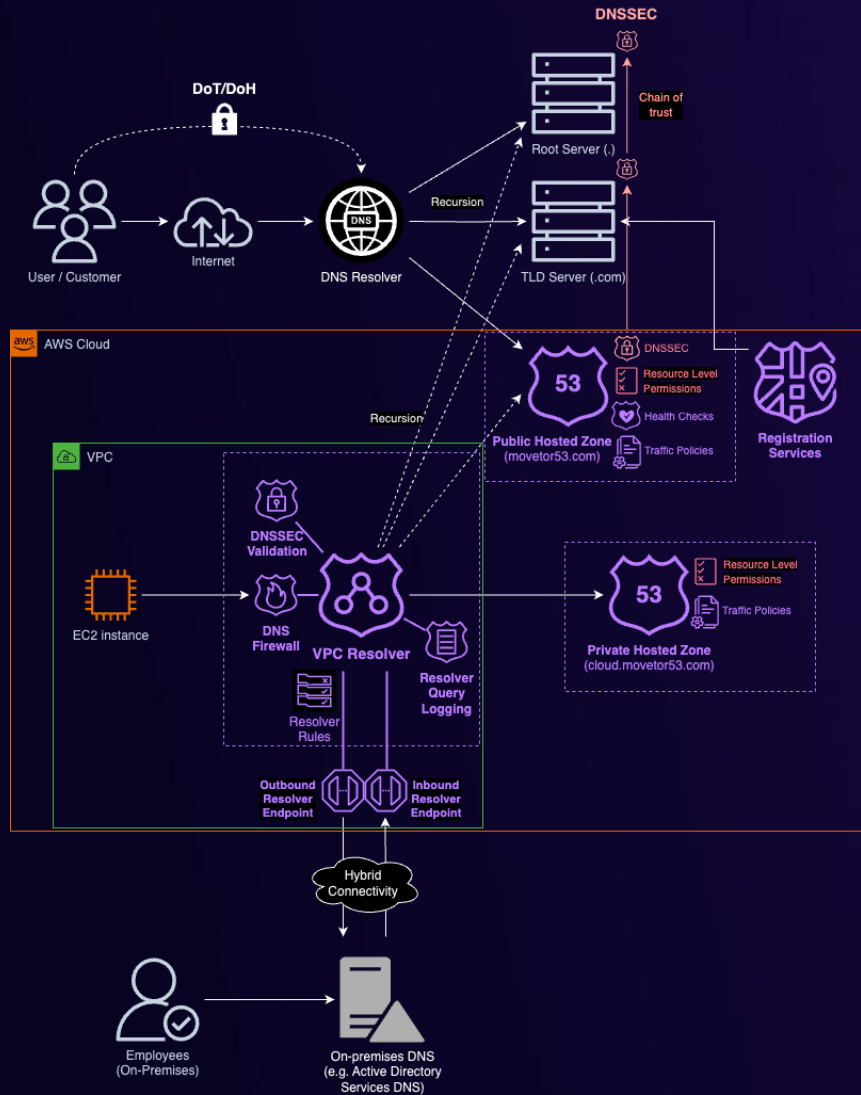
Use case 8 : Secure my public hosted zone

Use case 9: Troubleshoot DNS queries

Use case 10: DNS resiliency across providers

Ask us anything!

Amazon Route 53 use cases



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Thank you!

Steve Seymour

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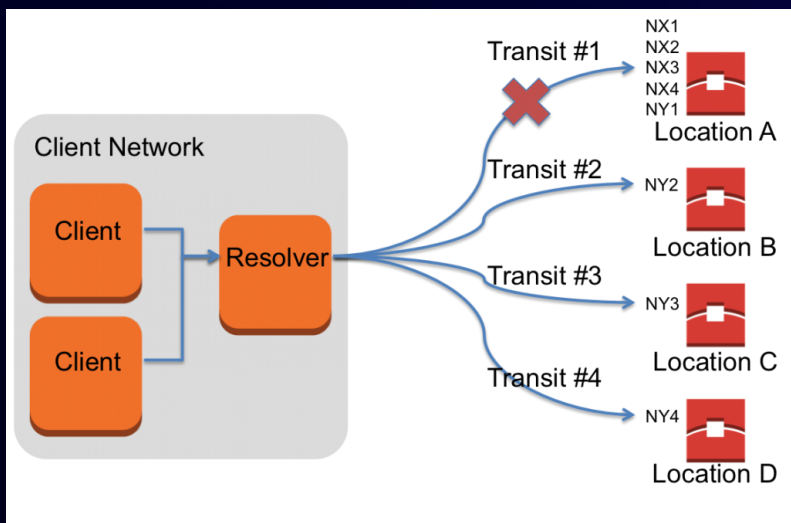
Christian Elsen

github.com/chriselsen



Please complete the session
survey in the mobile app

Why you don't need a multi-provider setup



AWS Architecture Blog

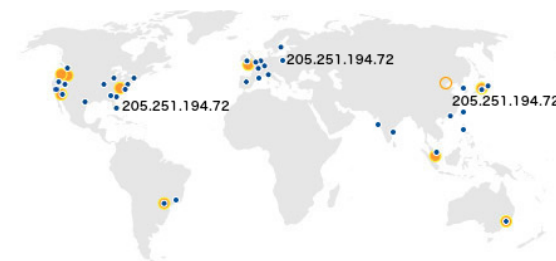
A Case Study in Global Fault Isolation

by Lee-Ming Zen | on 21 MAY 2014 | in [Amazon Route 53](#), [Architecture](#) | [Permalink](#) | [Share](#)

In a previous blog post, we talked about [using shuffle sharding to get magical fault isolation](#). Today, we'll examine a specific use case that Route 53 employs and one of the interesting tradeoffs we decided to make as part of our sharding. Then, we'll discuss how you can employ some of these concepts in your own applications.

Overview of Anycast DNS

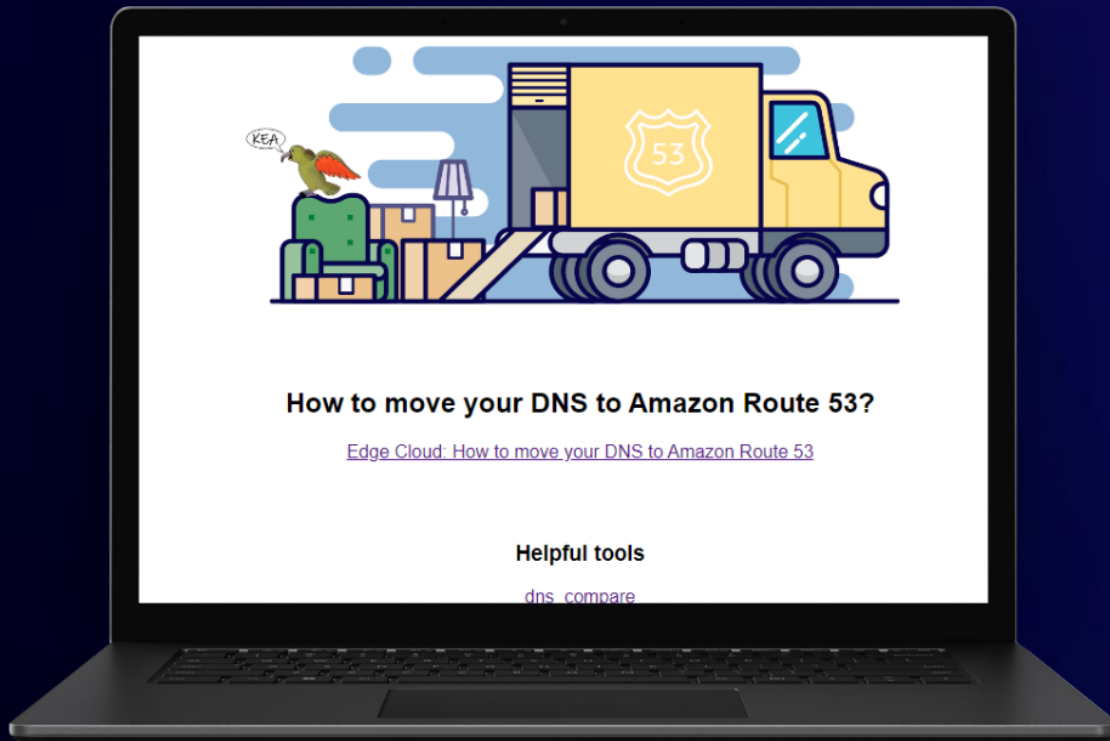
One of our goals at Amazon Route 53 is to provide low-latency DNS resolution to customers. We do this, in part, by announcing our IP addresses using "anycast" from [over 50 edge locations](#) around the globe. Anycast works by routing packets to the closest (network-wise) location that is "advertising" a particular address. In the image below, we can see that there are three locations, all of which can receive traffic for the 205.251.194.72 address.



(Blue circles represent edge locations; orange circles represent AWS regions)



<https://www.movetor53.com>



Amazon Route 53 global edge network

GLOBAL NETWORK

Redundant 400 GbE network and private capacity between all Regions except for the AWS China*

DIRECT CONNECT

Connect to every AWS Region from over 130 AWS Direct Connect locations worldwide (excluding AWS China Regions*)

EDGE NETWORKING

600+ PoPs in 50 countries and 100+ cities, with direct peering to all major ISPs

- KEY**
- Edge location
 - Multiple edge locations

